Algebraic structures of the Lindblad equation

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Abstract

This Mathematica notebook accompanies the paper The Algebraic Structure of the Lindblad Equation. The notebook illustrates the proposed algebraic formulation through the example of a driven transmon qubit described by the Lindblad master equation.

The example is organized as a step-by-step implementation of the algebraic method. It begins by constructing the Hermitian matrix basis and the associated algebraic objects, followed by the recursive generation of the operators required to represent the Liouville superoperator. The model-dependent quantities are then assembled to obtain the complete Liouville superoperator, which is compared with the one obtained by the conventional direct construction, demonstrating their exact agreement.

The notebook is fully parameterized with respect to the number of qubits, allowing the same implementation to be extended to larger Hilbert spaces. Although some symbolic calculations become computationally demanding as the system size increases, the code provides a general framework that can be readily adapted to study a broad class of finite-dimensional open quantum systems.

Licence

LindbladAlgebra Mathematica examples accompanying the paper

“The Algebraic Structure of the Lindblad Equation”

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In this section we create a series of Mathematica functions that facilitate the computation of the matrix basis, structure constants and Liouville superoperator.

Functions

In this section we create a series of Mathematica functions that facilitate the computation of the matrix basis, structure constants and Liouville superoperator.

Functions

Example 1. One qubit.

One-qubit matrix basis h

In the following example we verify the main relations derived in the paper for the one qubit case.

In[•]:=
```
nq = 1;
```

We begin by constructing the one-qubit basis $\mathbf{h}_n$ and its associated structure constant superoperators $\boldsymbol{Z}$, $\boldsymbol{Z}^*$ and $\boldsymbol{X}$.

In[•]:=
```
h = QubitBasis[nq];
{b, c} = BCStructureConstantsFromQubits[nq];
z = ZStructureConstantsFromQubits[nq];
zt = Conjugate[z];
x = XBasisFromQubits[nq];
```

The complex structure-constant superoperators can be visualized with the following line of code.

```
In[•]:= MatrixForm[z[[3]]]
```

Out[•]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & -i\sqrt{2} \\ \sqrt{2} & 0 & 0 & 0 \\ 0 & i\sqrt{2} & 0 & 0 \end{pmatrix}$$

It can be verified that $\boldsymbol{Z}$ and $\boldsymbol{Z}^*$ constitute to independent realizations of the Lie algebra associated with h by showing that the following difference, obtained by subtracting the right-hand side from the left-hand side of Eqs. (69) and (70), vanish identically.

```
In[•]:= zz = Transpose[z.Transpose[z], {1, 3, 2, 4}];
        commu = SparseArray[zz - Transpose[zz]];
        PossibleZeroQ[SparseArray[commu + 2 i c.z]]
```

Out[•]=

SparseArray[Specified elements: 0 Dimensions: {4, 4, 4, 4} Default: True]

```
In[•]:= ztzt = Transpose[zt.Transpose[zt], {1, 3, 2, 4}];
        commu = SparseArray[ztzt - Transpose[ztzt]];
        PossibleZeroQ[SparseArray[commu - 2 i c.zt]]
```

Out[•]=

SparseArray[Specified elements: 0 Dimensions: {4, 4, 4, 4} Default: True]

Similarly we can verify that $\boldsymbol{Z}$, $\boldsymbol{Z}^*$ commute.

```
In[•]:= zzt = Transpose[z.ConjugateTranspose[z]];
        ztz = Transpose[zt.Transpose[z], {3, 1, 2, 4}];
        commu = SparseArray[zzt - ztz];
        PossibleZeroQ[commu]
```

Out[•]=

SparseArray[Specified elements: 0 Dimensions: {4, 4, 4, 4} Default: True]

The structure of the 16 elements of $\boldsymbol{X}$ can be visualized using the following line of code.

```
In[•]:= MatrixForm[x[[2, 3]]]
```

Out[•]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 \end{pmatrix}$$

The Lie and Jordan algebraic relations of $\boldsymbol{X}$ can be verified by evaluating the following differences,

obtained by subtracting the right-hand side from the left-hand side of Eqs. (81) and (82), and verifying that they vanish identically. This calculation may exceed the available memory if $n_q \geq 2$.

```
In[•]:= xx = Transpose[x.Transpose[x, {2, 3, 1, 4}], {1, 2, 5, 3, 4, 6}];
commu = SparseArray[xx - Transpose[xx, {3, 4, 1, 2, 5, 6}]];
anticommu = SparseArray[xx + Transpose[xx, {3, 4, 1, 2, 5, 6}]];
```

```
In[•]:= PossibleZeroQ[SparseArray[commu -
    1/4 Transpose[(zt.Transpose[x, {1, 4, 2, 3}].z - z.Transpose[x, {1, 4, 2, 3}].zt),
      {1, 3, 5, 6, 2, 4}]]]
```

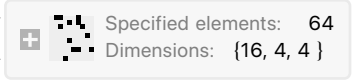
```
Out[•]= SparseArray[ 
                     Specified elements: 0
                     Dimensions: {4, 4, 4, 4, 4, 4}
                     Default: True ]
```

```
In[•]:= PossibleZeroQ[SparseArray[anticommu -
    1/4 Transpose[(zt.Transpose[x, {1, 4, 2, 3}].z + z.Transpose[x, {1, 4, 2, 3}].zt),
      {1, 3, 5, 6, 2, 4}]]]
```

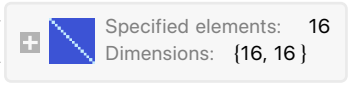
```
Out[•]= SparseArray[ 
                     Specified elements: 0
                     Dimensions: {4, 4, 4, 4, 4, 4}
                     Default: True ]
```

The orthonormality of $\mathfrak{X}_{n^2}$ is verified by flattening the first two indices of $\boldsymbol{X}$ into a single index and then evaluating the Hilbert-Schmidt inner product

```
In[•]:= X = Flatten[x, 1]
```

```
Out[•]= SparseArray[ 
                     Specified elements: 64
                     Dimensions: {16, 4, 4}
                     ]
```

```
In[•]:= XX = SparseArray[TensorContract[X.Transpose[X], {{2, 4}}]]
DiagonalMatrixQ[XX]
```

```
Out[•]= SparseArray[ 
                     Specified elements: 16
                     Dimensions: {16, 16}
                     ]
```

```
Out[•]= True
```

Hamiltonian

A convenient system for illustrating the results developed in this work is the transmon qubit. Since the number of quantum levels retained in the model can be chosen arbitrarily, it provides a simple framework for studying the behavior of the algebraic structures introduced above as the Hilbert-space

dimension increases. We first define the lowering and rasing operators as well as the jump operators

```
In[ ]:= m = 2^nq;
        n = 2^(2 nq);
        a = SparseArray[Table[{n + 1, n + 2} → Sqrt[n + 1], {n, 0, m - 2}], m];
        at = SparseArray[Table[{n + 1, n} → Sqrt[n], {n, 1, m - 1}], m];
        A = SparseArray[Table[{n + 1, n + 1, n + 2} → Sqrt[n + 1], {n, 0, m - 2}], {m - 1, m, m}];
        At = SparseArray[Table[{n + 1, n + 2, n + 1} → Sqrt[n + 1], {n, 0, m - 2}], {m - 1, m, m}];
```

The Hamiltonian in the interaction picture, given by Eq. (122) is given by

```
In[ ]:= ψ = Exp[-i (Δ t + ϕ)] Map[Exp, i α t (1 - Range[m - 1])];
        ψt = Exp[i (Δ t + ϕ)] Map[Exp, -i α t (1 - Range[m - 1])];
        Hi = Ω/2 (ψ.A + ψt.At);
        MatrixForm[Hi]
```

```
Out[ ]//MatrixForm=
( 0                          (1/2) E^(-i (t Δ+ϕ)) Ω
  (1/2) E^(i (t Δ+ϕ)) Ω      0                     )
```

The Hamiltonian vector is obtained by projecting the Hamiltonian on the elements of the set $\mathbf{h}_n$ through Eq. (23).

```
In[ ]:= H = ExpToTrig[ProjectOperatorOntoBasis[Hi, h]];
        MatrixForm[H]
```

```
Out[ ]//MatrixForm=
( 0
  Ω Cos[t Δ+ϕ]/Sqrt[2]
  Ω Sin[t Δ+ϕ]/Sqrt[2]
  0                    )
```

Lindbladian

To construct the Lindbladian, we first introduce the corresponding jump operators. In the case of transmons, two decoherence processes are particularly relevant: energy relaxation and pure dephasing. Energy relaxation describes the loss of an excitation from the transmon to its environment and is represented by the Lindblad jump operator $J_1 = a$. This process is characterized by the relaxation time T_1, time or equivalently by the relaxation rate $\Gamma_1 = 1/T_1$. Although the corresponding contribution to the Lindbladian is time independent in the Schrödinger picture, it becomes time dependent in the interaction picture used here to express the Hamiltonian. For a harmonic oscillator a simple scalar phase cancels completely, but for the anharmonic transmon the phase contains the number operator $n=a^\dagger a$, that introduces the time dependency. This time dependency is easily introduced through the phases and decomposing into transition operators the lowering and rising operators

```
In[•]:= J1 = SparseArray[Table[{n, n, n + 1} → √n Exp[-i α n t], {n, 1, m - 1}], {m - 1, m, m}];
        J1t = SparseArray[Table[{n, n + 1, n} → √n Exp[i α n t], {n, 1, m - 1}], {m - 1, m, m}];
```

The Lindbladian corresponding to the energy relaxation part is defined by Eq. (38)

```
In[•]:= γ1 = SparseArray[Flatten[Table[{ii, jj} → γ10, {ii, m - 1}, {jj, m - 1}]], {m - 1, m - 1}];
        Γ1 = GammaFromgamma[h, γ1, J1, J1t];
        MatrixForm[Γ1]
```

Out[•]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{\gamma 10}{4} & \frac{i\,\gamma 10}{4} & 0 \\ 0 & -\frac{i\,\gamma 10}{4} & \frac{\gamma 10}{4} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Pure dephasing describes a phenomenon that does not change the energy-level populations but instead, destroys coherence between different energy eigenstates. It is characterized by the jump operator $J_2 = a^\dagger a$

```
In[•]:= J2 = SparseArray[{at.a}];
        J2t = SparseArray[{at.a}];
```

and the pure-dephasing time T_2 or rate $\Gamma_2 = 1/T_1 - 1/2\,T_2$ which is associated

```
In[•]:= γ2 = SparseArray[{{γ20}}];
        Γ2 = GammaFromgamma[h, γ2, J2, J2t];
        MatrixForm[Γ2]
```

Out[•]//MatrixForm=

$$\begin{pmatrix} \frac{\gamma 20}{4} & 0 & 0 & -\frac{\gamma 20}{4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\frac{\gamma 20}{4} & 0 & 0 & \frac{\gamma 20}{4} \end{pmatrix}$$

The full Lindbladian is therefore

```
In[•]:= Γ = Γ1 + Γ2;
        MatrixForm[Γ]
```

Out[•]//MatrixForm=

$$\begin{pmatrix} \frac{\gamma 20}{4} & 0 & 0 & -\frac{\gamma 20}{4} \\ 0 & \frac{\gamma 10}{4} & \frac{i\,\gamma 10}{4} & 0 \\ 0 & -\frac{i\,\gamma 10}{4} & \frac{\gamma 10}{4} & 0 \\ -\frac{\gamma 20}{4} & 0 & 0 & \frac{\gamma 20}{4} \end{pmatrix}$$

Algebraic Method

Having obtained the quantities required to construct the Hamiltonian and Lindbladian contributions to the Liouville superoperator, namely ***H*** and Γ, we can proceed to compute Λ using Eq. (60), yielding

```
In[ ]:= Λ = SparseArray[Simplify[Normal[LambdaFromHamiltonianLindbladian[z, H, Γ]]]];
        MatrixForm[Λ]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} \frac{1}{4}(-2\gamma 10-\gamma 20) & -i\Omega\,\text{Cos}[t\Delta+\phi] & -i\Omega\,\text{Sin}[t\Delta+\phi] & \frac{\gamma 10}{4} \\ i\Omega\,\text{Cos}[t\Delta+\phi] & \frac{\gamma 10}{4} & \frac{i\gamma 10}{4} & 0 \\ i\Omega\,\text{Sin}[t\Delta+\phi] & -\frac{i\gamma 10}{4} & \frac{\gamma 10}{4} & 0 \\ \frac{\gamma 10}{4} & 0 & 0 & \frac{\gamma 20}{4} \end{pmatrix}$$

The Liouville superoperator is obtained using Eq. (59).

```
In[ ]:= Lam = SparseArray[Simplify[Normal[LiouvilleSOFromLamda[x, Λ]]]];
        MatrixForm[Lam]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4}(-\gamma 10-\gamma 20) & 0 & \Omega\,\text{Sin}[t\Delta+\phi] \\ 0 & 0 & \frac{1}{4}(-\gamma 10-\gamma 20) & -\Omega\,\text{Cos}[t\Delta+\phi] \\ \frac{\gamma 10}{2} & -\Omega\,\text{Sin}[t\Delta+\phi] & \Omega\,\text{Cos}[t\Delta+\phi] & -\frac{\gamma 10}{2} \end{pmatrix}$$

Direct method

The Liouville superoperator is also computed in the same matrix basis for comparison. The two methods are found to yield identical results.

```
In[ ]:= Ldm = SparseArray[Simplify[Normal[LiouvilleSOFromProjection[h, H, Γ]]]];
        MatrixForm[Ldm]
```

Out[ ]//MatrixForm=

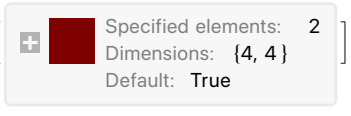
$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4}(-\gamma 10-\gamma 20) & 0 & \Omega\,\text{Sin}[t\Delta+\phi] \\ 0 & 0 & \frac{1}{4}(-\gamma 10-\gamma 20) & -\Omega\,\text{Cos}[t\Delta+\phi] \\ \frac{\gamma 10}{2} & -\Omega\,\text{Sin}[t\Delta+\phi] & \Omega\,\text{Cos}[t\Delta+\phi] & -\frac{\gamma 10}{2} \end{pmatrix}$$

Comparison of the algebraic method and the direct method

To compare the Liouville superoperators obtained using the algebraic and direct methods, we subtract one superoperator from the other.

```
In[ ]:= PossibleZeroQ[SparseArray[Lam - Ldm]]
```

Out[ ]=

```
SparseArray[ 
                Specified elements: 2
                Dimensions: {4, 4}
                Default: True ]
```

Liouville equation for the dynamical map

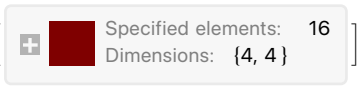
To verify that the dynamical differential equations for the coordinate vector, Eq. (112), are equivalent to those derived from the Lindbladian, we directly project the Liouville operator acting on the dynami-

cal map, as given on the right-hand side of Eq. (110), onto the basis ***X***. We then subtract the resulting expression from the right-hand side of Eq. (112), which should yield zero if both formulations are equivalent. First, we define the elements of the dynamical map as in Eq. (107).

```
In[•]:= V = SparseArray[Array[v[##][t] &, {n, n}]];
```

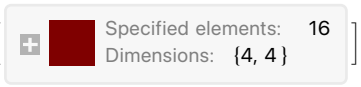
We calculate the right-hand side of the dynamical differential equations using the algebraically derived Eq. (112).

```
In[•]:= DiffEqsAlgebraicMethod =
  SparseArray[Flatten[V].Flatten[Transpose[SparseArray[1/4 zt.Λ.z], {1, 3, 2, 4}], 1]]
```

```
Out[•]=
SparseArray[ 
  Specified elements: 16
  Dimensions: {4, 4}
]
```

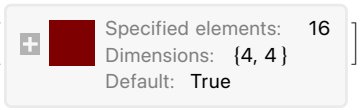
Here, we project the Liouville operator obtained through the direct method, acting on the dynamical map as given on the right-hand side of Eq. (110), onto the basis ***X***.

```
In[•]:= DiffEqsDirectMethod = TensorContract[
   Transpose[V].TensorContract[x.Transpose[x, {2, 3, 1, 4}].Ldm, {{3, 6}}], {{1, 2}}]
```

```
Out[•]=
SparseArray[ 
  Specified elements: 16
  Dimensions: {4, 4}
]
```

Finally, we compare the results obtained by both methods.

```
In[•]:= PossibleZeroQ[SparseArray[DiffEqsDirectMethod - DiffEqsAlgebraicMethod]]
```

```
Out[•]=
SparseArray[ 
  Specified elements: 16
  Dimensions: {4, 4}
  Default: True
]
```

Example 2. Two qubits.

Two-qubit matrix basis h

In the following example we verify the main relations derived in the paper for the two qubit case.

```
In[•]:= nq = 2;
```

We begin by constructing the two-qubit basis $\mathbf{h}_n$ and its associated structure constant superoperators ***Z***, $\boldsymbol{Z}^*$ and ***X***.

```
In[•]:= h = QubitBasis[nq];
        {b, c} = BCStructureConstantsFromQubits[nq];
        z = ZStructureConstantsFromQubits[nq];
        zt = Conjugate[z];
        x = XBasisFromQubits[nq];
```

The complex structure-constant superoperators can be visualized with the following line of code.

```
In[•]:= MatrixForm[z[[3]]]
```

Out[•]//MatrixForm=

$$\begin{pmatrix}
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0
\end{pmatrix}$$

It can be verified that $\boldsymbol{Z}$ and $\boldsymbol{Z}^*$ constitute to independent realizations of the Lie algebra associated with h by showing that the following difference, obtained by subtracting the right-hand side from the left-hand side of Eqs. (69) and (70), vanish identically.

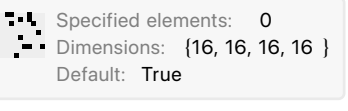
```
In[•]:= zz = Transpose[z.Transpose[z], {1, 3, 2, 4}];
        commu = SparseArray[zz - Transpose[zz]];
        PossibleZeroQ[SparseArray[commu + 2 i c.z]]
```

Out[•]=

```
SparseArray[ 
                Specified elements: 0
                Dimensions: {16, 16, 16, 16 }
                Default: True ]
```

```
In[•]:= ztzt = Transpose[zt.Transpose[zt], {1, 3, 2, 4}];
        commu = SparseArray[ztzt - Transpose[ztzt]];
        PossibleZeroQ[SparseArray[commu - 2 i c.zt]]
```

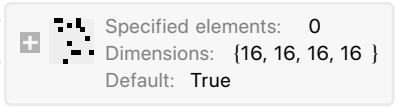
Out[•]=

```
SparseArray[ 
                Specified elements: 0
                Dimensions: {16, 16, 16, 16 }
                Default: True ]
```

Similarly we can verify that $\boldsymbol{Z}$, $\boldsymbol{Z}^*$ commute.

```
In[•]:= zzt = Transpose[z.ConjugateTranspose[z]];
        ztz = Transpose[zt.Transpose[z], {3, 1, 2, 4}];
        commu = SparseArray[zzt - ztz];
        PossibleZeroQ[commu]
```

```
Out[•]=
```

SparseArray[]

The structure of the 16 elements of $\boldsymbol{X}$ can be visualized using the following line of code.

```
In[•]:= MatrixForm[x[[2, 3]]]
```

```
Out[•]//MatrixForm=
```

$$\begin{pmatrix}
0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0
\end{pmatrix}$$

The Lie and Jordan algebraic relations of $\boldsymbol{X}$ can be verified by evaluating the following differences, obtained by subtracting the right-hand side from the left-hand side of Eqs. (81) and (82), and verifying that they vanish identically. This calculation may exceed the available memory if $n_q \geq 2$.

```
In[•]:= xx = Transpose[x.Transpose[x, {2, 3, 1, 4}], {1, 2, 5, 3, 4, 6}];
        commu = SparseArray[xx - Transpose[xx, {3, 4, 1, 2, 5, 6}]];
        anticommu = SparseArray[xx + Transpose[xx, {3, 4, 1, 2, 5, 6}]];
```

```
In[ ]:= PossibleZeroQ[SparseArray[commu -
          1/4 Transpose[(zt.Transpose[x, {1, 4, 2, 3}].z - z.Transpose[x, {1, 4, 2, 3}].zt),
            {1, 3, 5, 6, 2, 4}]]]
```

```
Out[ ]=
SparseArray[  Specified elements: 0
                                                               Dimensions: {16, 16, 16, 16, 16, 16 }
                                                               Default: True ]
```

```
In[ ]:= PossibleZeroQ[SparseArray[anticommu -
          1/4 Transpose[(zt.Transpose[x, {1, 4, 2, 3}].z + z.Transpose[x, {1, 4, 2, 3}].zt),
            {1, 3, 5, 6, 2, 4}]]]
```

```
Out[ ]=
SparseArray[  Specified elements: 0
                                                               Dimensions: {16, 16, 16, 16, 16, 16 }
                                                               Default: True ]
```

The orthonormality of $\mathfrak{X}_{n^2}$ is verified by flattening the first two indices of $\boldsymbol{X}$ into a single index and then evaluating the Hilbert-Schmidt inner product

```
In[ ]:= X = Flatten[x, 1]
```

```
Out[ ]=
SparseArray[  Specified elements : 4096
                                                               Dimensions: {256, 16, 16 } ]
```

```
In[ ]:= XX = SparseArray[TensorContract[X.Transpose[X], {{2, 4}}]]
         DiagonalMatrixQ[XX]
```

```
Out[ ]=
SparseArray[  Specified elements: 256
                                                               Dimensions: {256, 256 } ]
```

```
Out[ ]=
True
```

Hamiltonian

A convenient system for illustrating the results developed in this work is the transmon qubit. Since the number of quantum levels retained in the model can be chosen arbitrarily, it provides a simple framework for studying the behavior of the algebraic structures introduced above as the Hilbert-space dimension increases. We first define the lowering and rasing operators as well as the jump operators

```
In[ ]:= m = 2^nq;
        n = 2^(2 nq);
        a = SparseArray[Table[{n + 1, n + 2} → Sqrt[n + 1], {n, 0, m - 2}], m];
        at = SparseArray[Table[{n + 1, n} → Sqrt[n], {n, 1, m - 1}], m];
        A = SparseArray[Table[{n + 1, n + 1, n + 2} → Sqrt[n + 1], {n, 0, m - 2}], {m - 1, m, m}];
        At = SparseArray[Table[{n + 1, n + 2, n + 1} → Sqrt[n + 1], {n, 0, m - 2}], {m - 1, m, m}];
```

The Hamiltonian in the interaction picture, given by Eq. (122) is given by

```
In[ ]:= ψ = Exp[-i (Δ t + ϕ)] Map[Exp, i α t (1 - Range[m - 1])];
        ψt = Exp[i (Δ t + ϕ)] Map[Exp, -i α t (1 - Range[m - 1])];
        Hi = Ω/2 (ψ.A + ψt.At);
        MatrixForm[Hi]
```

Out[ ]//MatrixForm=

$$\begin{pmatrix} 0 & \frac{1}{2} e^{-i (t\Delta+\phi)} \Omega & 0 & 0 \\ \frac{1}{2} e^{i (t\Delta+\phi)} \Omega & 0 & \frac{e^{-i t\alpha - i (t\Delta+\phi)} \Omega}{\sqrt{2}} & 0 \\ 0 & \frac{e^{i t\alpha + i (t\Delta+\phi)} \Omega}{\sqrt{2}} & 0 & \frac{1}{2} \sqrt{3} e^{-2 i t\alpha - i (t\Delta+\phi)} \Omega \\ 0 & 0 & \frac{1}{2} \sqrt{3} e^{2 i t\alpha + i (t\Delta+\phi)} \Omega & 0 \end{pmatrix}$$

The Hamiltonian vector is obtained by projecting the Hamiltonian on the elements of the set $\mathbf{h}_n$ through Eq. (23).

```
In[•]:= H = ExpToTrig[ProjectOperatorOntoBasis[Hi, h]];
        MatrixForm[H]
```

Out[•]//MatrixForm=

$$\begin{pmatrix} 0 \\ \frac{1}{2}\,\Omega\,\text{Cos}[t\,\Delta+\phi]+\frac{1}{2}\,\sqrt{3}\,\Omega\,\text{Cos}[2\,t\,\alpha+t\,\Delta+\phi] \\ \frac{1}{2}\,\Omega\,\text{Sin}[t\,\Delta+\phi]+\frac{1}{2}\,\sqrt{3}\,\Omega\,\text{Sin}[2\,t\,\alpha+t\,\Delta+\phi] \\ 0 \\ 0 \\ \frac{\Omega\,\text{Cos}[t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ -\frac{\Omega\,\text{Sin}[t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{\Omega\,\text{Sin}[t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ \frac{\Omega\,\text{Cos}[t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{2}\,\Omega\,\text{Cos}[t\,\Delta+\phi]-\frac{1}{2}\,\sqrt{3}\,\Omega\,\text{Cos}[2\,t\,\alpha+t\,\Delta+\phi] \\ \frac{1}{2}\,\Omega\,\text{Sin}[t\,\Delta+\phi]-\frac{1}{2}\,\sqrt{3}\,\Omega\,\text{Sin}[2\,t\,\alpha+t\,\Delta+\phi] \\ 0 \end{pmatrix}$$

Lindbladian

To construct the Lindbladian, we first introduce the corresponding jump operators. In the case of transmons, two decoherence processes are particularly relevant: energy relaxation and pure dephasing. Energy relaxation describes the loss of an excitation from the transmon to its environment and is represented by the Lindblad jump operator $J_1 = a$. This process is characterized by the relaxation time T_1, time or equivalently by the relaxation rate $\Gamma_1 = 1/T_1$. Although the corresponding contribution to the Lindbladian is time independent in the Schrödinger picture, it becomes time dependent in the interaction picture used here to express the Hamiltonian. For a harmonic oscillator a simple scalar phase cancels completely, but for the anharmonic transmon the phase contains the number operator n=$a^\dagger$ a, that introduces the time dependency. This time dependency is easily introduced through the phases and decomposing into transition operators the lowering and rising operators

```
In[•]:= J1 = SparseArray[Table[{n, n, n + 1} → √n Exp[-i α n t], {n, 1, m - 1}], {m - 1, m, m}];
        J1t = SparseArray[Table[{n, n + 1, n} → √n Exp[i α n t], {n, 1, m - 1}], {m - 1, m, m}];
```

The Lindbladian corresponding to the energy relaxation part is defined by Eq. (38)

```
In[•]:= γ1 = SparseArray[Flatten[Table[{ii, jj} → γ10, {ii, m - 1}, {jj, m - 1}]], {m - 1, m - 1}];
        Γ1 = GammaFromgamma[h, γ1, J1, J1t];
        MatrixForm[Γ1]
```

Out[•]//MatrixForm=

$$\begin{pmatrix}
0 & & & & & & & & & & & & & & & \\
0 & \frac{1}{2}\left(\frac{1}{2}\,e^{-i\,t\,\alpha}\left(\frac{1}{2}\,e^{i\,t\,\alpha}\,\gamma 10+\frac{1}{2}\,\sqrt{3}\,e^{3\,i\,t\,\alpha}\,\gamma 10\right)+\frac{1}{2}\,\sqrt{3}\,e^{-3\,i\,t\,\alpha}\left(\frac{1}{2}\,e^{i\,t\,\alpha}\,\gamma 10+\frac{1}{2}\,\sqrt{3}\,e^{3\,i\,t\,\alpha}\,\gamma 10\right)\right) & & & & & & & & & & & & & \\
0 & \frac{1}{2}\left(\frac{1}{2}\,e^{-i\,t\,\alpha}\left(-\frac{1}{2}\,i\,e^{i\,t\,\alpha}\,\gamma 10-\frac{1}{2}\,i\,\sqrt{3}\,e^{3\,i\,t\,\alpha}\,\gamma 10\right)+\frac{1}{2}\,\sqrt{3}\,e^{-3\,i\,t\,\alpha}\left(-\frac{1}{2}\,i\,e^{i\,t\,\alpha}\,\gamma 10-\frac{1}{2}\,i\,\sqrt{3}\,e^{3\,i\,t\,\alpha}\,\gamma\right. & & & & & & & & & & & & & \\
0 & & 0 & & & & & & & & & & & & & \\
0 & & & 0 & & & & & & & & & & & & \\
0 & & & & \frac{1}{2}\left(\frac{1}{2}\,\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\,\gamma 10+\frac{e^{i\,t\,\alpha}\,\gamma 10}{2\,\sqrt{2}}\right) & & & & & & & & & & & \\
0 & & & & & \frac{1}{2}\left(\frac{1}{2}\,i\,\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\,\gamma 10+\frac{i\,e^{i\,t\,\alpha}\,\gamma 10}{2\,\sqrt{2}}\right) & & & & & & & & & \\
0 & & & & & & 0 & & & & & & & & & \\
0 & & & & & & & 0 & & & & & & & & \\
0 & & & & & & & & \frac{1}{2}\left(-\frac{1}{2}\,i\,\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\,\gamma 10-\frac{i\,e^{i\,t\,\alpha}\,\gamma 10}{2\,\sqrt{2}}\right) & & & & & & & \\
0 & & & & & & & & & \frac{1}{2}\left(\frac{1}{2}\,\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\,\gamma 10+\frac{e^{i\,t\,\alpha}\,\gamma 10}{2\,\sqrt{2}}\right) & & & & & & \\
0 & & & & & & & & & & 0 & & & & & \\
0 & & & & & & & & & & & 0 & & & & \\
0 & \frac{1}{2}\left(\frac{1}{2}\,e^{-i\,t\,\alpha}\left(\frac{1}{2}\,e^{i\,t\,\alpha}\,\gamma 10-\frac{1}{2}\,\sqrt{3}\,e^{3\,i\,t\,\alpha}\,\gamma 10\right)+\frac{1}{2}\,\sqrt{3}\,e^{-3\,i\,t\,\alpha}\left(\frac{1}{2}\,e^{i\,t\,\alpha}\,\gamma 10-\frac{1}{2}\,\sqrt{3}\,e^{3\,i\,t\,\alpha}\,\gamma 10\right)\right) & & & & & & & & & & & & & \\
0 & \frac{1}{2}\left(\frac{1}{2}\,e^{-i\,t\,\alpha}\left(-\frac{1}{2}\,i\,e^{i\,t\,\alpha}\,\gamma 10+\frac{1}{2}\,i\,\sqrt{3}\,e^{3\,i\,t\,\alpha}\,\gamma 10\right)+\frac{1}{2}\,\sqrt{3}\,e^{-3\,i\,t\,\alpha}\left(-\frac{1}{2}\,i\,e^{i\,t\,\alpha}\,\gamma 10+\frac{1}{2}\,i\,\sqrt{3}\,e^{3\,i\,t\,\alpha}\,\gamma\right. & & & & & & & & & & & & & \\
0 & & 0 & & & & & & & & & & & & &
\end{pmatrix}$$

Pure dephasing describes a phenomenon that does not change the energy-level populations but instead, destroys coherence between different energy eigenstates. It is characterized by the jump operator $J_2 = a^\dagger a$

```
In[•]:= J2 = SparseArray[{at.a}];
        J2t = SparseArray[{at.a}];
```

and the pure-dephasing time T_2 or rate $\Gamma_2 = 4/T_1 - 2/T_2$ which is associated

In[•]:=
```
ϒ2 = SparseArray[{{ϒ20}}]
Γ2 = GammaFromgamma[h, ϒ2, J2, J2t];
MatrixForm[Γ2]
```

Out[•]=

SparseArray [Specified elements: 1 Dimensions: {1, 1}]

Out[•]//MatrixForm=

$$\begin{pmatrix}
\frac{9\,\gamma 20}{2} & 0 & 0 & -\frac{3\,\gamma 20}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3\,\gamma 20 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\,\gamma 20}{2} & 0 & 0 & \frac{\gamma 20}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \gamma 20 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-3\,\gamma 20 & 0 & 0 & \gamma 20 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\,\gamma 20 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

The full Lindbladian is therefore

```
In[•]:= Γ = SparseArray[Simplify[Normal[Γ1 + Γ2]]];
        MatrixForm[Γ]
```

Out[•]//MatrixForm=

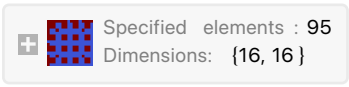
$$\begin{pmatrix}
\frac{9\gamma 20}{2} & 0 & 0 \\
0 & \frac{1}{8}\left(4+\sqrt{3}\,e^{-2it\alpha}+\sqrt{3}\,e^{2it\alpha}\right)\gamma 10 & \frac{1}{8}\,i\,e^{-2it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\left(1+\sqrt{3}\,e^{2it\alpha}\right) \\
0 & -\frac{1}{8}\,i\,e^{-2it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\left(1+\sqrt{3}\,e^{2it\alpha}\right)\gamma 10 & \frac{1}{8}\left(4+\sqrt{3}\,e^{-2it\alpha}+\sqrt{3}\,e^{2it\alpha}\right)\gamma 10 \\
-\frac{3\gamma 20}{2} & 0 & 0 \\
0 & 0 & 0 \\
0 & \frac{e^{-it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\gamma 10}{4\sqrt{2}} & \frac{i\,e^{-it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\gamma 10}{4\sqrt{2}} \\
0 & \frac{i\,e^{-it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\gamma 10}{4\sqrt{2}} & -\frac{e^{-it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\gamma 10}{4\sqrt{2}} \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & -\frac{i\,e^{-it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\gamma 10}{4\sqrt{2}} & \frac{e^{-it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\gamma 10}{4\sqrt{2}} \\
0 & \frac{e^{-it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\gamma 10}{4\sqrt{2}} & \frac{i\,e^{-it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\gamma 10}{4\sqrt{2}} \\
0 & 0 & 0 \\
-3\gamma 20 & 0 & 0 \\
0 & \frac{1}{8}\left(-2+\sqrt{3}\,e^{-2it\alpha}-\sqrt{3}\,e^{2it\alpha}\right)\gamma 10 & -\frac{1}{8}\,i\,e^{-2it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\left(-1+\sqrt{3}\,e^{2it\alpha}\right) \\
0 & \frac{1}{8}\,i\,e^{-2it\alpha}\left(\sqrt{3}+e^{2it\alpha}\right)\left(-1+\sqrt{3}\,e^{2it\alpha}\right)\gamma 10 & \frac{1}{8}\left(-2+\sqrt{3}\,e^{-2it\alpha}-\sqrt{3}\,e^{2it\alpha}\right)\gamma 10 \\
0 & 0 & 0
\end{pmatrix}$$

Algebraic Method

Having obtained the quantities required to construct the Hamiltonian and Lindbladian contributions to the Liouville superoperator, namely ***H*** and Γ, we can proceed to compute Λ using Eq. (60), yielding

In[•]:=
```
Λ = LambdaFromHamiltonianLindbladian[z, H, Γ]
MatrixForm[Simplify[Normal[Λ]]]
```

Out[•]=

SparseArray[]

Out[•]//MatrixForm=

$$\begin{pmatrix}
-3\,\gamma 10 - \frac{5\,\gamma 20}{2} & -i\,\Omega\left(\text{Cos}[t\,\Delta+\phi] + \sqrt{3}\,\text{Cos}[t\,(2\,\alpha+\Delta)+\phi]\right) & -i\,\Omega \\
i\,\Omega\left(\text{Cos}[t\,\Delta+\phi] + \sqrt{3}\,\text{Cos}[t\,(2\,\alpha+\Delta)+\phi]\right) & \frac{1}{8}\left(4+\sqrt{3}\,e^{-2\,i\,t\,\alpha}+\sqrt{3}\,e^{2\,i\,t\,\alpha}\right)\gamma 10 & \frac{1}{8}\,i \\
i\,\Omega\left(\text{Sin}[t\,\Delta+\phi] + \sqrt{3}\,\text{Sin}[t\,(2\,\alpha+\Delta)+\phi]\right) & -\frac{1}{8}\,i\,e^{-2\,i\,t\,\alpha}\left(\sqrt{3}+e^{2\,i\,t\,\alpha}\right)\left(1+\sqrt{3}\,e^{2\,i\,t\,\alpha}\right)\gamma 10 & \\
\frac{\gamma 10}{2} & 0 & \\
0 & 0 & \\
i\,\sqrt{2}\,\Omega\,\text{Cos}[t\,(\alpha+\Delta)+\phi] & \frac{e^{-i\,t\,\alpha}\left(\sqrt{3}+e^{2\,i\,t\,\alpha}\right)\gamma 10}{4\,\sqrt{2}} & \\
-i\,\sqrt{2}\,\Omega\,\text{Sin}[t\,(\alpha+\Delta)+\phi] & \frac{i\,e^{-i\,t\,\alpha}\left(\sqrt{3}+e^{2\,i\,t\,\alpha}\right)\gamma 10}{4\,\sqrt{2}} & \\
0 & 0 & \\
0 & 0 & \\
i\,\sqrt{2}\,\Omega\,\text{Sin}[t\,(\alpha+\Delta)+\phi] & -\frac{i\,e^{-i\,t\,\alpha}\left(\sqrt{3}+e^{2\,i\,t\,\alpha}\right)\gamma 10}{4\,\sqrt{2}} & \\
i\,\sqrt{2}\,\Omega\,\text{Cos}[t\,(\alpha+\Delta)+\phi] & \frac{e^{-i\,t\,\alpha}\left(\sqrt{3}+e^{2\,i\,t\,\alpha}\right)\gamma 10}{4\,\sqrt{2}} & \\
0 & 0 & \\
\gamma 10 & 0 & \\
i\,\Omega\left(\text{Cos}[t\,\Delta+\phi] - \sqrt{3}\,\text{Cos}[t\,(2\,\alpha+\Delta)+\phi]\right) & \frac{1}{8}\left(-2+\sqrt{3}\,e^{-2\,i\,t\,\alpha}-\sqrt{3}\,e^{2\,i\,t\,\alpha}\right)\gamma 10 & -\frac{1}{8}\,i \\
i\,\Omega\left(\text{Sin}[t\,\Delta+\phi] - \sqrt{3}\,\text{Sin}[t\,(2\,\alpha+\Delta)+\phi]\right) & \frac{1}{8}\,i\,e^{-2\,i\,t\,\alpha}\left(\sqrt{3}+e^{2\,i\,t\,\alpha}\right)\left(-1+\sqrt{3}\,e^{2\,i\,t\,\alpha}\right)\gamma 10 & \\
-\gamma 20 & 0 &
\end{pmatrix}$$

The Liouville superoperator is obtained using Eq. (59).

In[•]:=
```
Lam = LiouvilleSOFromLamda[x, Λ]
MatrixForm[Simplify[Normal[Lam]]]
```

Out[•]=

SparseArray[ Specified elements: 256 Dimensions: {16, 16}]

Out[•]//MatrixForm=

$$\begin{pmatrix}
0 & 0 & 0 \\
0 & \frac{1}{4}\left(-3\,\gamma 10-\gamma 20\right) & 0 \\
0 & 0 & \frac{1}{4}\left(-3\,\gamma 10-\gamma 20\right) \\
\frac{\gamma 10}{2} & -\frac{1}{2}\,\Omega\left(\text{Sin}[t\,\Delta+\phi]+\sqrt{3}\,\text{Sin}[t\,(2\,\alpha+\Delta)+\phi]\right) & \frac{1}{2}\,\Omega\left(\text{Cos}[t\,\Delta+\phi]+\sqrt{3}\,\text{Cos}[t\,(2\,\alpha+\Delta)+\phi]\right) \\
0 & 0 & 0 \\
0 & \frac{1}{4}\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(1+e^{2\,i\,t\,\alpha}\right)\gamma 10 & -\frac{1}{4}\,i\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(-1+e^{2\,i\,t\,\alpha}\right)\gamma 10 \\
0 & -\frac{1}{4}\,i\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(-1+e^{2\,i\,t\,\alpha}\right)\gamma 10 & -\frac{1}{4}\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(1+e^{2\,i\,t\,\alpha}\right)\gamma 10 \\
0 & \frac{\Omega\,\text{Sin}[t\,(\alpha+\Delta)+\phi]}{\sqrt{2}} & \frac{\Omega\,\text{Cos}[t\,(\alpha+\Delta)+\phi]}{\sqrt{2}} \\
0 & 0 & 0 \\
0 & \frac{1}{4}\,i\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(-1+e^{2\,i\,t\,\alpha}\right)\gamma 10 & \frac{1}{4}\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(1+e^{2\,i\,t\,\alpha}\right)\gamma 10 \\
0 & \frac{1}{4}\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(1+e^{2\,i\,t\,\alpha}\right)\gamma 10 & -\frac{1}{4}\,i\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(-1+e^{2\,i\,t\,\alpha}\right)\gamma 10 \\
0 & -\frac{\Omega\,\text{Cos}[t\,(\alpha+\Delta)+\phi]}{\sqrt{2}} & \frac{\Omega\,\text{Sin}[t\,(\alpha+\Delta)+\phi]}{\sqrt{2}} \\
\frac{\gamma 10}{2} & 0 & 0 \\
0 & \frac{\gamma 10}{2} & 0 \\
0 & 0 & \frac{\gamma 10}{2} \\
-\frac{\gamma 10}{2} & -\frac{1}{2}\,\Omega\left(\text{Sin}[t\,\Delta+\phi]-\sqrt{3}\,\text{Sin}[t\,(2\,\alpha+\Delta)+\phi]\right) & \frac{1}{2}\,\Omega\left(\text{Cos}[t\,\Delta+\phi]-\sqrt{3}\,\text{Cos}[t\,(2\,\alpha+\Delta)+\phi]\right)
\end{pmatrix}$$

Direct method

The Liouville superoperator is also computed in the same matrix basis for comparison. The two methods are found to yield identical results.

In[•]:=
```
Ldm = LiouvilleSOFromProjection[h, H, Γ]
MatrixForm[Simplify[Normal[Ldm]]]
```

Out[•]=

SparseArray [ Specified elements: 256
Dimensions: {16, 16}]

Out[•]//MatrixForm=

$$\begin{pmatrix}
0 & 0 & 0 \\
0 & \frac{1}{4}\left(-3\,\gamma 10-\gamma 20\right) & 0 \\
0 & 0 & \frac{1}{4}\left(-3\,\gamma 10-\gamma 20\right) \\
\frac{\gamma 10}{2} & -\frac{1}{2}\,\Omega\left(\text{Sin}[t\,\Delta+\phi]+\sqrt{3}\,\text{Sin}[t\,(2\,\alpha+\Delta)+\phi]\right) & \frac{1}{2}\,\Omega\left(\text{Cos}[t\,\Delta+\phi]+\sqrt{3}\,\text{Cos}[t\,(2\,\alpha+\Delta)+\phi]\right) \\
0 & 0 & 0 \\
0 & \frac{1}{4}\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(1+e^{2\,i\,t\,\alpha}\right)\gamma 10 & -\frac{1}{4}\,i\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(-1+e^{2\,i\,t\,\alpha}\right)\gamma 10 \\
0 & -\frac{1}{4}\,i\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(-1+e^{2\,i\,t\,\alpha}\right)\gamma 10 & -\frac{1}{4}\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(1+e^{2\,i\,t\,\alpha}\right)\gamma 10 \\
0 & \frac{\Omega\,\text{Sin}[t\,(\alpha+\Delta)+\phi]}{\sqrt{2}} & \frac{\Omega\,\text{Cos}[t\,(\alpha+\Delta)+\phi]}{\sqrt{2}} \\
0 & 0 & 0 \\
0 & \frac{1}{4}\,i\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(-1+e^{2\,i\,t\,\alpha}\right)\gamma 10 & \frac{1}{4}\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(1+e^{2\,i\,t\,\alpha}\right)\gamma 10 \\
0 & \frac{1}{4}\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(1+e^{2\,i\,t\,\alpha}\right)\gamma 10 & -\frac{1}{4}\,i\sqrt{\frac{3}{2}}\,e^{-i\,t\,\alpha}\left(-1+e^{2\,i\,t\,\alpha}\right)\gamma 10 \\
0 & -\frac{\Omega\,\text{Cos}[t\,(\alpha+\Delta)+\phi]}{\sqrt{2}} & \frac{\Omega\,\text{Sin}[t\,(\alpha+\Delta)+\phi]}{\sqrt{2}} \\
\frac{\gamma 10}{2} & 0 & 0 \\
0 & \frac{\gamma 10}{2} & 0 \\
0 & 0 & \frac{\gamma 10}{2} \\
-\frac{\gamma 10}{2} & -\frac{1}{2}\,\Omega\left(\text{Sin}[t\,\Delta+\phi]-\sqrt{3}\,\text{Sin}[t\,(2\,\alpha+\Delta)+\phi]\right) & \frac{1}{2}\,\Omega\left(\text{Cos}[t\,\Delta+\phi]-\sqrt{3}\,\text{Cos}[t\,(2\,\alpha+\Delta)+\phi]\right)
\end{pmatrix}$$

Comparison of the algebraic method and the direct method

To compare the Liouville superoperators obtained using the algebraic and direct methods, we subtract one superoperator from the other.

In[•]:=
```
PossibleZeroQ[SparseArray[Lam - Ldm]]
```

Out[•]=

SparseArray [ Specified elements: 96
Dimensions: {16, 16}
Default: True]

Liouville equation for the dynamical map

To verify that the dynamical differential equations for the coordinate vector, Eq. (112), are equivalent to those derived from the Lindbladian, we directly project the Liouville operator acting on the dynami-

cal map, as given on the right-hand side of Eq. (110), onto the basis $\boldsymbol{X}$. We then subtract the resulting expression from the right-hand side of Eq. (112), which should yield zero if both formulations are equivalent. First, we define the elements of the dynamical map as in Eq. (107).

```
In[•]:= V = SparseArray[Array[v[##][t] &, {n, n}]];
        fV = Flatten[V];
        {b, c} = BCStructureConstantsFromQubits[nq];
        z = ZStructureConstantsFromQubits[nq];
        zt = Conjugate[z];
```

We calculate the right-hand side of the dynamical differential equations using the algebraically derived Eq. (112).

```
In[•]:= DiffEqsAlgebraicMethod =
          SparseArray[fV.Flatten[Transpose[SparseArray[1/4 zt.Λ.z], {1, 3, 2, 4}], 1]]
```

```
Out[•]= SparseArray[ Specified elements: 256
                     Dimensions: {16, 16}
                     Data not saved. Save now ]
```

Here, we project the Liouville operator obtained through the direct method, acting on the dynamical map as given on the right-hand side of Eq. (110), onto the basis $\boldsymbol{X}$.

```
In[•]:= DiffEqsDirectMethod = TensorContract[
          Transpose[V].TensorContract[x.Transpose[x, {2, 3, 1, 4}].Ldm, {{3, 6}}], {{1, 2}}]
```

```
Out[•]= SparseArray[ Specified elements: 256
                     Dimensions: {16, 16}
                     Data not saved. Save now ]
```

Finally, we compare the results obtained by both methods.

```
In[•]:= PossibleZeroQ[SparseArray[DiffEqsDirectMethod - DiffEqsAlgebraicMethod]]
```

```
Out[•]= SparseArray[ Specified elements: 256
                     Dimensions: {16, 16}
                     Default: True ]
```

Example 3. Four-level transmon dynamics: *X*-gate implementation and leakage

To illustrate how to solve the resulting system of ordinary differential equations, we study the dynamics of a transmon truncated to its four lowest energy levels, using a pulse designed to implement an *X* (*NOT*) gate. This allows us to examine population leakage from the computational subspace, spanned

by $|0\rangle$ and $|1\rangle$, into the higher levels $|2\rangle$ and $|3\rangle$ during the gate operation. For this purpose we use the differential equations derived in the previous example (stored in the variable DiffEqsAlgebraicMethod). We flatten the time derivative of the dynamical map *V(t)* and equate it to the flattened right-hand side obtained above. This process yields the equations in the format required by NDSolve.

```
In[•]:= eqs = Thread[D[fV, t] == Flatten[DiffEqsAlgebraicMethod]];
```

We choose parameter values typical of transmon qubits: $\Omega_0/2\pi$ = 50MHz, $\alpha/2\pi$ = -300MHz, $\omega/2\pi$ = 3 GHz, T_1 = 150 μs and T_2 = 100 μs.

```
In[•]:= Ω0 = 2 π 0.05;
        α0 = -2 π 0.3;
        ω0 = 2 π 3.0;
        T1 = 150.0 × 10^3;
        T2 = 100.0 × 10^3;
        Γ10 = 2 / T1;
        Γ20 = 4/T2 - 2/T1;
```

Additionally, to facilitate comparison with known results, we use a Gaussian pulse with width $\sigma = 2\pi/\Omega_0$ and choose the phase ϕ to implement an X gate. We then construct the system of equations, stored in eqsn, using these parameter values.

```
In[•]:= σ = 2 π / Ω0;
        t0 = 4 σ;
        tmin = 0;
        tmax = 8 σ;
        Δ0 = 0.0;
        ϕ0 = 0;
        constants =
          {α → α0, Δ → Δ0, Ω → Ω0/(2 √(2 π)) Exp[-1/2 ((t - t0)/σ)^2], ϕ → ϕ0, ɣ10 → Γ10, ɣ20 → Γ20};
        eqsn = Chop[eqs /. constants];
```

We impose the initial condition *V(0)=I* . In terms of the expansion coefficients, this corresponds to $V_{i,j} = 0$ for *(i,j)* ≠ *(1,1)* and $V_{1,1}$= 4, since $X_{1,1}$=I/4 for nq=2

```
In[•]:= initialcond = Thread[(Normal[fV] /. {t → 0}) == SparseArray[{1} → 4, n^2]];
```

We combine the differential equations with the initial conditions and solve the resulting initial-value problem using NDSolve.

```
In[•]:= diffeqsn = Join[eqsn, initialcond];
        sol = NDSolve[diffeqsn, Normal[fV], {t, tmin, tmax}][[1]];
```

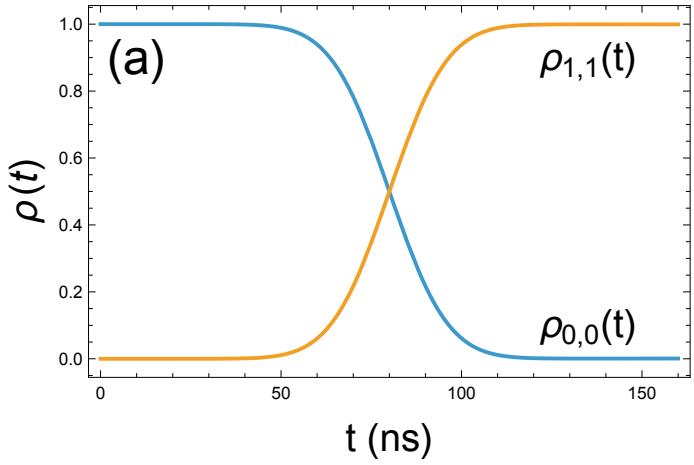
To obtain a particular case, we can set the transmon to be initially in the first state. Then, the density matrix can be expanded through Eq. (16) and the actual density matrix is recovered as expanding it in terms of the basis $\hbar_n$. We first try the case where the lowest level is fully populated and the remaining ones are empty

In[•]:=
```
ρ0 = ProjectOperatorOntoBasis[SparseArray[{1, 1} → 1, {m, m}], h];
ρ = fV.X.ρ0;
rho = Normal[ρ.h] /. sol;
```

Plotting the populations $\rho_{1,1}(t)$ and $\rho_{2,2}(t)$ of the two first quantum levels of the transmon as a function of time

In[•]:=
```
Show[
 Plot[{Re[rho[[1, 1]]], Re[rho[[2, 2]]]}, {t, tmin, tmax}, Axes → False,
  Frame → True, FrameLabel → {Style["t (ns)", 20], Style["ρ(t)", Italic, 20]}],
 Graphics[{
   Text[Style["ρ1,1(t)", 20], {135, 0.9}],
   Text[Style["ρ0,0(t)", 20], {135, 0.1}],
   Text[Style["(a)", 25], {10, 0.9}]
  }]
]
```

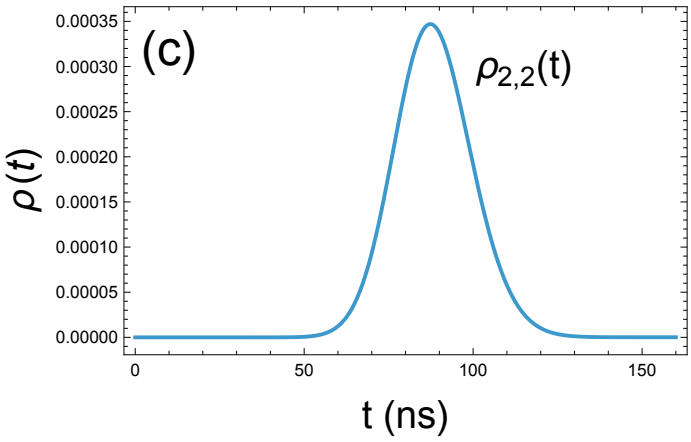
Out[•]=



We can also get a notion of the magnitude of the leakages of probability by plotting the density matrix $\rho_{3,3}$ and $\rho_{4,4}$

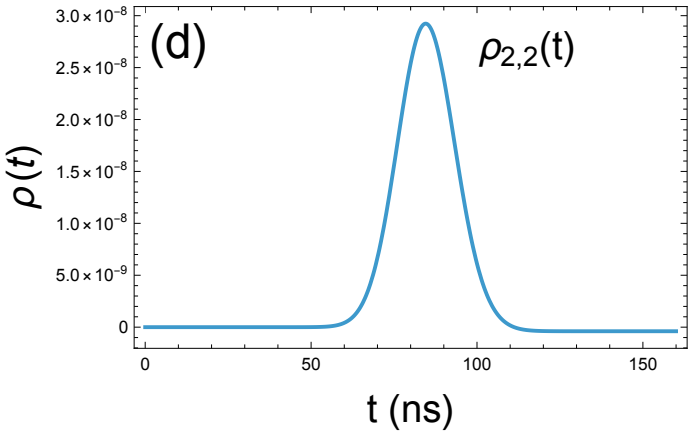
In[•]:=
```
Show[
 Plot[Re[rho[[3, 3]]], {t, tmin, tmax}, Axes → False, Frame → True,
  FrameLabel → {Style["t (ns)", 20], Style["ρ(t)", Italic, 20]}],
 Graphics[{
   Text[Style["ρ2,2(t)", 20], {115, 0.0003}],
   Text[Style["(c)", 25], {10, 0.00032}]
  }]
]
```

Out[•]=



In[•]:=
```
Show[
 Plot[Re[rho[[4, 4]]], {t, tmin, tmax}, Axes → False, Frame → True, PlotRange → All,
  FrameLabel → {Style["t (ns)", 20], Style["ρ(t)", Italic, 20]}],
 Graphics[{
   Text[Style["ρ2,2(t)", 20], {115, 2.7 × 10^-8}],
   Text[Style["(d)", 25], {10, 2.75 × 10^-8}]
  }]
]
```

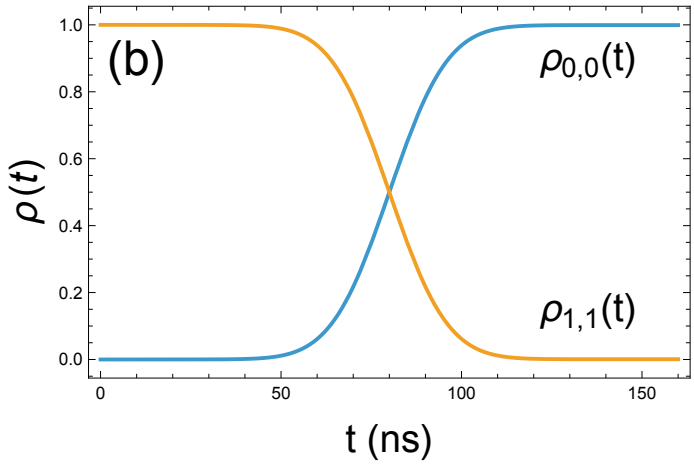
Out[•]=



We can also try other initial configurations, as one in which the second level is initially fully populated.

```
In[•]:= ρ0 = ProjectOperatorOntoBasis[SparseArray[{2, 2} → 1, {m, m}], h];
        ρ = fV.X.ρ0;
        rho = Normal[ρ.h] /. sol;
        Show[
         Plot[{Re[rho[[1, 1]]], Re[rho[[2, 2]]]}, {t, tmin, tmax}, Axes → False,
          Frame → True, FrameLabel → {Style["t (ns)", 20], Style["ρ(t)", Italic, 20]}],
         Graphics[{
           Text[Style["ρ1,1(t)", 20], {135, 0.15}],
           Text[Style["ρ0,0(t)", 20], {135, 0.9}],
           Text[Style["(b)", 25], {10, 0.9}]
          }]
        ]
```

Out[•]=



The preceding results suggest behavior consistent with an X gate. To assess how closely the evolution approximates this gate, we compute the dynamical map projected onto the computational subspace spanned by the two lowest energy levels, $|0\rangle$ and $|1\rangle$. We start by computing the dynamical mat in the Liouville space and evaluate it at t = tmax.

```
In[•]:= Vmat = Normal[fV.X] /. sol /. {t → tmax};
```

To obtain the dynamical map on the computational subspace spanned by the two lowest energy levels, $|0\rangle$ and $|1\rangle$ we construct a one-qubit operator basis, embed it in the four-level operator space, and project *V(t)* onto it.

```
In[•]:= g0 = QubitBasis[1];
        g = KroneckerProduct[SparseArray[{1, 1, 1} → 1, {1, 2, 2}], g0];
        T = TensorContract[g.Transpose[h], {{2, 4}}];
        MatrixForm[Chop[T.Vmat.Transpose[T], 1 × 10^-5]]
```

Out[•]//MatrixForm=

$$\begin{pmatrix} 1. & 0 & 0 & 0 \\ 0 & 0.998011 & 0 & -0.0279055 \\ -0.000280683 & 0 & -0.998498 & -0.000492926 \\ 0 & -0.0279055 & 0.000493025 & -0.998446 \end{pmatrix}$$

This result can be compared with the actual dynamical map of the gate *X*.

```
In[•]:= U = PauliMatrix[1];
        TensorContract[(g0.U).Transpose[g0.ConjugateTranspose[U]], {{2, 4}}] // MatrixForm
```

Out[•]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

This result indicates that the implemented operation approximates an *X* gate.

Example 4. Three qubits.

Three-qubit matrix basis h

In the following example we verify the main relations derived in the paper for the two qubit case.

```
In[•]:= nq = 3;
```

We begin by constructing the one-qubit basis $\mathbf{h}_n$ and its associated structure constant superoperators $\boldsymbol{Z}$, $\boldsymbol{Z}^*$ and $\boldsymbol{X}$.

```
In[•]:= h = QubitBasis[nq];
        {b, c} = BCStructureConstantsFromQubits[nq];
        z = ZStructureConstantsFromQubits[nq];
        zt = Conjugate[z];
        x = XBasisFromQubits[nq];
```

The complex structure-constant superoperators can be visualized with the following line of code.

```
In[•]:= MatrixForm[z[[3]]]
```

Out[•]//MatrixForm=

$$\begin{pmatrix}
0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

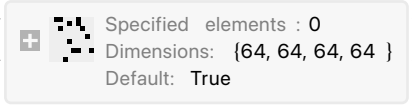
$$\begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{\sqrt{2}} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} \\
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{\sqrt{2}} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

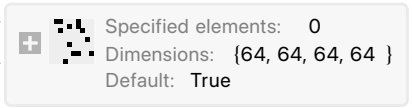
It can be verified that $\boldsymbol{Z}$ and $\boldsymbol{Z}^*$ constitute to independent realizations of the Lie algebra associated with h by showing that the following difference, obtained by subtracting the right-hand side from the left-hand side of Eqs. (69) and (70), vanish identically.

In[•]:=
```
zz = Transpose[z.Transpose[z], {1, 3, 2, 4}];
commu = SparseArray[zz - Transpose[zz]];
PossibleZeroQ[SparseArray[commu + 2 ⅈ c.z]]
```

Out[•]=

SparseArray[]

```
In[•]:= ztzt = Transpose[zt.Transpose[zt], {1, 3, 2, 4}];
        commu = SparseArray[ztzt - Transpose[ztzt]];
        PossibleZeroQ[SparseArray[commu - 2 i c.zt]]
```

```
Out[•]=
SparseArray[ 
  Specified elements: 0
  Dimensions: {64, 64, 64, 64 }
  Default: True ]
```

Similarly we can verify that $\boldsymbol{Z}$, $\boldsymbol{Z}^*$ commute.

```
In[•]:= zzt = Transpose[z.ConjugateTranspose[z]];
        ztz = Transpose[zt.Transpose[z], {3, 1, 2, 4}];
        commu = SparseArray[zzt - ztz];
        PossibleZeroQ[commu]
```

```
Out[•]=
SparseArray[ 
  Specified elements: 0
  Dimensions: {64, 64, 64, 64 }
  Default: True ]
```

The structure of the elements of $\boldsymbol{X}$ can be visualized using the following line of code.

```
In[•]:= MatrixForm[x[[2, 3]]]
```

```
Out[•]//MatrixForm=
```

$$\begin{pmatrix}
0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

$$\begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
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\end{pmatrix}$$

$$\begin{pmatrix}
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\end{pmatrix}$$

For some computations it is convenient to flatten the first two indices of X.

In[•]:=
```
X = Flatten[x, 1]
```

Out[•]=

SparseArray [Specified elements : 262 144
Dimensions: {4096, 64, 64 }]

Data not saved. Save now

Hamiltonian

A convenient system for illustrating the results developed in this work is the transmon qubit. Since the number of quantum levels retained in the model can be chosen arbitrarily, it provides a simple framework for studying the behavior of the algebraic structures introduced above as the Hilbert-space dimension increases. We first define the lowering and rasing operators as well as the jump operators

In[•]:=
```
m = 2^nq;
n = 2^2 nq;
a = SparseArray[Table[{n + 1, n + 2} → Sqrt[n + 1], {n, 0, m - 2}], m];
at = SparseArray[Table[{n + 1, n} → Sqrt[n], {n, 1, m - 1}], m];
A = SparseArray[Table[{n + 1, n + 1, n + 2} → Sqrt[n + 1], {n, 0, m - 2}], {m - 1, m, m}];
At = SparseArray[Table[{n + 1, n + 2, n + 1} → Sqrt[n + 1], {n, 0, m - 2}], {m - 1, m, m}];
```

The Hamiltonian in the interaction picture, given by Eq. (122) is given by

```
ψ = Exp[-i (Δ t + ϕ)] Map[Exp, i α t (1 - Range[m - 1])];
ψt = Exp[i (Δ t + ϕ)] Map[Exp, -i α t (1 - Range[m - 1])];
Hi = Ω/2 (ψ.A + ψt.At);
MatrixForm[Hi]
```

Out[•]//MatrixForm=

$$\begin{pmatrix}
0 & \frac{1}{2} e^{-i (t\Delta+\phi)} \Omega & 0 & 0 & 0 \\
\frac{1}{2} e^{i (t\Delta+\phi)} \Omega & 0 & \frac{e^{-i t\alpha - i (t\Delta+\phi)} \Omega}{\sqrt{2}} & 0 & 0 \\
0 & \frac{e^{i t\alpha + i (t\Delta+\phi)} \Omega}{\sqrt{2}} & 0 & \frac{1}{2}\sqrt{3}\, e^{-2 i t\alpha - i (t\Delta+\phi)} \Omega & 0 \\
0 & 0 & \frac{1}{2}\sqrt{3}\, e^{2 i t\alpha + i (t\Delta+\phi)} \Omega & 0 & e^{-3 i t\alpha - i (t\Delta+\phi)} \Omega \\
0 & 0 & 0 & e^{3 i t\alpha + i (t\Delta+\phi)} \Omega & 0 \\
0 & 0 & 0 & 0 & \frac{1}{2}\sqrt{5}\, e^{4 i t\alpha + i (t\Delta+\phi)} \Omega \\
0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

The Hamiltonian vector is obtained by projecting the Hamiltonian on the elements of the set $\mathbf{h}_n$ through Eq. (23).

In[•]:=
```
H = ExpToTrig[ProjectOperatorOntoBasis[Hi, h]];
MatrixForm[H]
```

Out[•]//MatrixForm=

$$\begin{pmatrix}
0 \\
\frac{\Omega \cos[t\Delta+\phi]}{2\sqrt{2}} + \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \cos[2t\alpha + t\Delta + \phi] + \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \cos[4t\alpha + t\Delta + \phi] + \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \cos[6t\alpha + t\Delta + \phi] \\
\frac{\Omega \sin[t\Delta+\phi]}{2\sqrt{2}} + \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \sin[2t\alpha + t\Delta + \phi] + \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \sin[4t\alpha + t\Delta + \phi] + \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \sin[6t\alpha + t\Delta + \phi] \\
0 \\
0 \\
\frac{1}{2}\Omega \cos[t\alpha + t\Delta + \phi] + \frac{1}{2}\sqrt{3}\,\Omega \cos[5t\alpha + t\Delta + \phi] \\
-\frac{1}{2}\Omega \sin[t\alpha + t\Delta + \phi] - \frac{1}{2}\sqrt{3}\,\Omega \sin[5t\alpha + t\Delta + \phi] \\
0 \\
0 \\
\frac{1}{2}\Omega \sin[t\alpha + t\Delta + \phi] + \frac{1}{2}\sqrt{3}\,\Omega \sin[5t\alpha + t\Delta + \phi] \\
\frac{1}{2}\Omega \cos[t\alpha + t\Delta + \phi] + \frac{1}{2}\sqrt{3}\,\Omega \cos[5t\alpha + t\Delta + \phi] \\
0 \\
0 \\
\frac{\Omega \cos[t\Delta+\phi]}{2\sqrt{2}} - \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \cos[2t\alpha + t\Delta + \phi] + \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \cos[4t\alpha + t\Delta + \phi] - \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \cos[6t\alpha + t\Delta + \phi] \\
\frac{\Omega \sin[t\Delta+\phi]}{2\sqrt{2}} - \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \sin[2t\alpha + t\Delta + \phi] + \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \sin[4t\alpha + t\Delta + \phi] - \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \sin[6t\alpha + t\Delta + \phi] \\
0 \\
0 \\
0 \\
0
\end{pmatrix}$$

$$\begin{pmatrix}
0 \\ 0 \\ \frac{\Omega \text{Cos}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ -\frac{\Omega \text{Sin}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ 0 \\ 0 \\ -\frac{\Omega \text{Sin}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ -\frac{\Omega \text{Cos}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{\Omega \text{Sin}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ \frac{\Omega \text{Cos}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{\Omega \text{Cos}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ -\frac{\Omega \text{Sin}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\
\frac{\Omega \text{Cos}[t\,\Delta+\phi]}{2\sqrt{2}} + \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Cos}[2\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Cos}[4\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Cos}[6\,t\,\alpha+t\,\Delta+\phi] \\
\frac{\Omega \text{Sin}[t\,\Delta+\phi]}{2\sqrt{2}} + \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Sin}[2\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Sin}[4\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Sin}[6\,t\,\alpha+t\,\Delta+\phi] \\
0 \\ 0 \\ \frac{1}{2}\,\Omega \text{Cos}[t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{3}\,\Omega \text{Cos}[5\,t\,\alpha+t\,\Delta+\phi] \\ -\frac{1}{2}\,\Omega \text{Sin}[t\,\alpha+t\,\Delta+\phi] + \frac{1}{2}\sqrt{3}\,\Omega \text{Sin}[5\,t\,\alpha+t\,\Delta+\phi] \\ 0 \\ 0 \\ \frac{1}{2}\,\Omega \text{Sin}[t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{3}\,\Omega \text{Sin}[5\,t\,\alpha+t\,\Delta+\phi] \\ \frac{1}{2}\,\Omega \text{Cos}[t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{3}\,\Omega \text{Cos}[5\,t\,\alpha+t\,\Delta+\phi] \\ 0 \\ 0 \\
\frac{\Omega \text{Cos}[t\,\Delta+\phi]}{2\sqrt{2}} - \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Cos}[2\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Cos}[4\,t\,\alpha+t\,\Delta+\phi] + \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Cos}[6\,t\,\alpha+t\,\Delta+\phi]
\end{pmatrix}$$

$$\begin{pmatrix} \frac{\Omega \text{Sin}[t\,\Delta+\phi]}{2\sqrt{2}} - \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Sin}[2\,t\,\alpha + t\,\Delta + \phi] - \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Sin}[4\,t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Sin}[6\,t\,\alpha + t\,\Delta + \phi \\ 0 \end{pmatrix}$$

Lindbladian

To construct the Lindbladian, we first introduce the corresponding jump operators. In the case of transmons, two decoherence processes are particularly relevant: energy relaxation and pure dephasing. Energy relaxation describes the loss of an excitation from the transmon to its environment and is represented by the Lindblad jump operator $J_1 = a$. This process is characterized by the relaxation time T_1, time or equivalently by the relaxation rate $\Gamma_1 = 1/T_1$. Although the corresponding contribution to the Lindbladian is time independent in the Schrödinger picture, it becomes time dependent in the interaction picture used here to express the Hamiltonian. For a harmonic oscillator a simple scalar phase cancels completely, but for the anharmonic transmon the phase contains the number operator n=$a^\dagger a$, that introduces the time dependency. This time dependency is easily introduced through the phases and decomposing into transition operators the lowering and rising operators

In[•]:=
```
J1 = SparseArray[Table[{n, n, n + 1} → √n Exp[-i α n t], {n, 1, m - 1}], {m - 1, m, m}];
J1t = SparseArray[Table[{n, n + 1, n} → √n Exp[i α n t], {n, 1, m - 1}], {m - 1, m, m}];
```

The Lindbladian corresponding to the energy relaxation part is defined by Eq. (38)

In[•]:=
```
γ1 = SparseArray[Flatten[Table[{ii, jj} → γ10, {ii, m - 1}, {jj, m - 1}]], {m - 1, m - 1}];
Γ1 = GammaFromgamma[h, γ1, J1, J1t];
MatrixForm[Γ1]
```

Out[•]//MatrixForm=

(… 1 …)

Full expression not available (original memory size: 1.9 MB)

Pure dephasing describes a phenomenon that does not change the energy-level populations but instead, destroys coherence between different energy eigenstates. It is characterized by the jump operator $J_2 = a^\dagger a$

In[•]:=
```
J2 = SparseArray[{at.a}];
J2t = SparseArray[{at.a}];
```

and the pure-dephasing time T_2 or rate $\Gamma_2 = 4\,T_1 - 2/T_2$ which is associated

In[•]:=
```
γ2 = SparseArray[{{γ20}}];
Γ2 = GammaFromgamma[h, γ2, J2, J2t];
MatrixForm[Γ2]
```

Out[•]//MatrixForm=

$$\begin{pmatrix} 49\,\gamma 20 & 0 & 0 & -7\,\gamma 20 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -14\,\gamma 20 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -7\,\gamma 20 & 0 & 0 & \gamma 20 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\,\gamma 20 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix}
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\end{pmatrix}$$

The full Lindbladian is therefore

In[•]:=
```
Γ = Γ1 + Γ2;
MatrixForm[Γ2]
```

Out[•]//MatrixForm=

$$\begin{pmatrix}
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Algebraic Method

Having obtained the quantities required to construct the Hamiltonian and Lindbladian contributions to the Liouville superoperator, namely ***H*** and Γ, we can proceed to compute Λ using Eq. (60), yielding

In[•]:=
```
Λ = LambdaFromHamiltonianLindbladian[z, H, Γ]
MatrixForm[Λ]
```

Out[•]=

SparseArray [Specified elements: 736
Dimensions: {64, 64}]

Data not saved. Save now

Out[•]//MatrixForm=

(… 1 …)

Full expression not available (original memory size: 5.4 MB)

The Liouville superoperator is obtained using Eq. (59).

In[•]:=
```
Lam = SparseArray[LiouvilleSOFromLamda[x, Λ]]
```

Out[•]=

SparseArray [Specified elements: 2816
Dimensions: {64, 64}]

Data not saved. Save now

Direct method

The Liouville superoperator is also computed in the same matrix basis for comparison. The two methods are found to yield identical results.

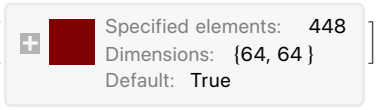
```
In[•]:= Ldm = SparseArray[LiouvilleSOFromProjection[h, H, Γ]];
```

Comparison of the algebraic method and the direct method

To compare the Liouville superoperators obtained using the algebraic and direct methods, we subtract one superoperator from the other.

```
In[•]:= PossibleZeroQ[SparseArray[Lam - Ldm]]
```

```
Out[•]=
```

SparseArray[]

Liouville equation for the dynamical map

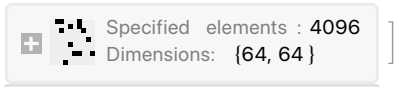
To verify that the dynamical differential equations for the coordinate vector, Eq. (112), are equivalent to those derived from the Lindbladian, we directly project the Liouville operator acting on the dynamical map, as given on the right-hand side of Eq. (110), onto the basis $\boldsymbol{X}$. We then subtract the resulting expression from the right-hand side of Eq. (112), which should yield zero if both formulations are equivalent. First, we define the elements of the dynamical map as in Eq. (107).

```
In[•]:= V = SparseArray[Array[v[##][t] &, {n, n}]];
```

We calculate the right-hand side of the dynamical differential equations using the algebraically derived Eq. (112).

```
In[•]:= DiffEqsAlgebraicMethod =
  SparseArray[Flatten[V].Flatten[Transpose[SparseArray[1/4 zt.Λ.z], {1, 3, 2, 4}], 1]]
```

```
Out[•]=
```

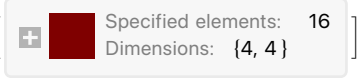
SparseArray[]

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Here, we project the Liouville operator obtained through the direct method, acting on the dynamical map as given on the right-hand side of Eq. (110), onto the basis $\boldsymbol{X}$.

```
In[•]:= DiffEqsDirectMethod = TensorContract[
    Transpose[V].TensorContract[x.Transpose[x, {2, 3, 1, 4}].Ldm, {{3, 6}}], {{1, 2}}]
```

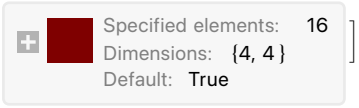
```
Out[•]=
```

SparseArray[]

Finally, we compare the results obtained by both methods.

In[•]:=
```
PossibleZeroQ[SparseArray[DiffEqsDirectMethod - DiffEqsAlgebraicMethod]]
```

Out[•]=

SparseArray []